Low Cost High Precision Animal Feeding and Smart Farm Monitoring System

Final Report – CENG424 Embedded Computer Systems

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***Abstract*—The current poultry industry depends heavily on regular feed delivery and checking environmental conditions. Standard approaches are labor-intensive, error prone, and don't allow for remote monitoring. Small to medium-sized chicken farm operators especially need an efficient, self-operating system to handle feeding tasks and environmental oversight when they're not present. This project developed an automated chicken feeding and environmental monitoring system using an ESP32 microcontroller, temperature sensing equipment, and a servo motor feeding system. The system functions independently and includes remote access through a mobile app that provides live monitoring, operational control, and immediate alert notifications—including when the application isn't actively running. A server system handles sensor information and user instructions while storing data permanently. The design was evaluated through step-by-step testing, checking communication, control functions, and alert systems both separately and as a complete unit. Early results indicate the system works correctly under testing conditions and offers a useful, affordable approach to enhance poultry farm operations, showing promise for practical farm use.**

***Index Terms*—Embedded systems, ESP32, smart agriculture, IoT automation, mobile control, poultry farming.**

I. Introduction

Modern poultry operations demand continuous oversight for feeding and environmental control. Manual approaches are inefficient and lack remote capabilities. This project presents a smart system tailored to small and mid-scale poultry farms, offering automation and remote access via a mobile interface. The goal is to reduce manual labor, improve reliability, and grant farmers remote oversight.

The system leverages an ESP32 microcontroller to manage a servo-powered auger for feed dispensing, and a DHT22 sensor for environmental tracking. A Node.js backend—paired with an SQLite database—coordinates communication and stores historical data. A Flutter mobile app allows users to monitor and control the system remotely, complete with real-time alerts.

Project goals include:

- Automate feeding based on timers or manual commands.
- Monitor temperature and humidity and notify users of threshold breaches.
- Offer a mobile interface for real-time remote access and notifications.
- Enable low-latency, reliable device-server-client communication.
- Record historical data for potential future analysis.

II. Literature Review

Recent innovations in smart farming integrate IoT and AI to streamline poultry management. Existing solutions offer automated control, data analytics, and remote monitoring.

Karun et al. [1] developed an IoT-based poultry system using various sensors and a mobile interface, prioritizing user-friendliness. However, their hardware is complex and may not be affordable for smaller farms. Our system simplifies the design with cost-effective ESP32 hardware.

Himel et al. [2] proposed an AI-enhanced farm system capable of learning optimal feeding times and identifying health risks through predictive modeling. While innovative, it's overengineered for farms with limited technical capacity. We focus instead on essential automation with minimal setup.

Our approach stands out through its simplicity, affordability, and accessibility—key for practical use in low-resource farming environments. Unlike other complex AI-based systems, our solution provides core functionality with an intuitive mobile interface.

A. Strengths in Existing Research

- Himel et al. use AI to improve feeding schedules and predict poultry health issues.

B. Limitations in Existing Research

- Karun et al.'s design requires expensive, complex hardware.
- Himel et al. assume access to advanced computational resources for AI deployment.

This project overcomes these issues by offering a simpler, affordable system suited to a wide range of farm sizes.

III. Methodology

The project combines embedded hardware, IoT protocols, backend development, and mobile UX to build a fully functional smart poultry solution. All components were designed modularly and tested independently before full integration.

A. System Design

- Three-tiered architecture: device (ESP32), backend server, and client app.
- TCP used for device-server communication; HTTP/WebSocket for app-server communication.
- Hardware: servo motor (feeder), DHT22 sensor (environment), ESP32, and SQLite.

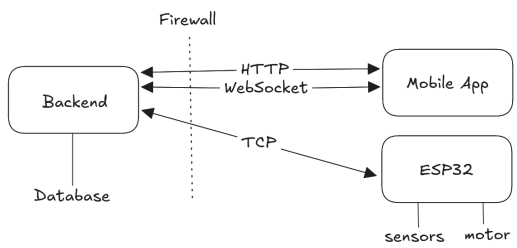

Fig. 1. System Design Architecture

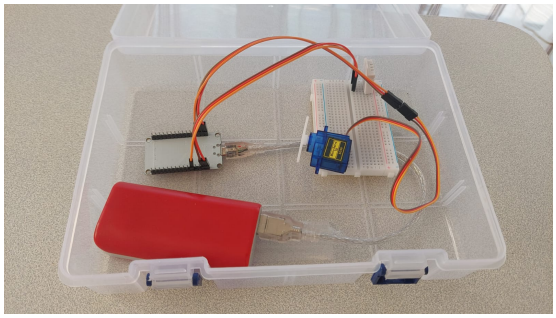

Fig. 2. ESP32 Circuit Elements

B. ESP32 Programming

- Wi-Fi connection using persistent TCP socket.
- Sensor data collection and motor control via PWM.
- Communication via JSON or custom-delimited strings.
- Supports periodic updates and threshold-triggered alerts.

C. Backend Implementation

- Node.js server supporting TCP, HTTP, and WebSocket protocols.
- RESTful APIs for system interaction and data queries.
- SQLite database stores logs and configurations.
- Commands from mobile app are relayed to ESP32 via TCP.

D. Flutter Mobile App

- Interface designed for ease of use.
- Fetches sensor data, sends feeding commands.
- Uses WebSocket for real-time interaction.
- Local notifications triggered based on thresholds.

Fig. 3. Home Screen

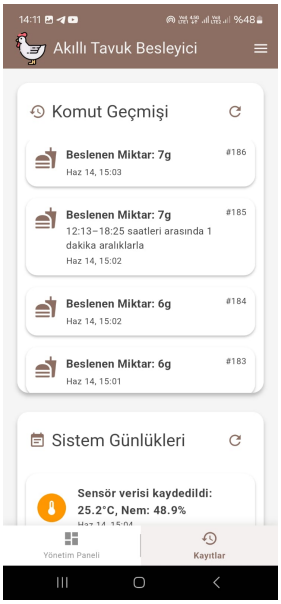

Fig. 4. Logs Screen

E. Integration and Testing

- ESP32-backend integration verified with test data.
- Mobile app tested for UI, data accuracy, and control latency.
- End-to-end scenarios simulated, including feed commands and environment changes.
- Network instability tests ensured graceful fallback behavior.

Fig. 5. Final Product

IV. Experimental Results

This section summarizes the main development-stage tests and their outcomes.

A. ESP32 Connectivity

1) Wi-Fi: Connected to Wi-Fi using ‘WiFi.h‘.
Result: Stable connection with DHCP.

2) HTTP: Tested GET/POST requests using ‘HTTP-Client.h‘.
Result: Successful, responses handled correctly.

3) WebSocket: Tested using existing ESP32 libraries.
Result: Unstable. Switched to raw TCP.

4) TCP: Established persistent connection with backend via raw sockets.
Result: Stable two-way communication.

B. ESP32 Hardware Testing

1) Servo: Controlled using ‘ESP32Servo.h‘ with PWM.
Result: Accurate feed mechanism.

2) DHT22 Sensor: Interfaced via GPIO using ‘DHT.h‘.
Result: Reliable temperature and humidity data.

C. Integration Testing

1) ESP32 to Backend: Live communication demoed with sensor and control updates.
Result: Real-time data transfer confirmed.

2) Feeding System: Auger mechanism driven by servo.
Result: Consistent feed dispensing.

D. Flutter Mobile App

1) UI: Responsive design with intuitive controls.
Result: Smooth navigation.

2) HTTP: Tested API calls using ‘http‘ package.
Result: Successful interaction with backend.

3) WebSocket: Live updates tested using ”web-socket-channel”.
Result: Works, minor latency noted.

4) Notifications: Implemented with ”flutter-local-notifications”.
Result: Correctly triggered alerts.

E. Backend Testing

1) API: Full CRUD verified via Postman.
Result: Functional endpoints.

2) Communication Protocols: All three protocols supported and tested.
Result: Stable except WebSocket under load.

3) Database: SQLite used for logging.
Result: Fast read/write.

V. Conclusion

The system successfully demonstrated smart poultry automation using low-cost hardware and scalable software. The ESP32, DHT22 sensor, and servo motor-driven auger feed mechanism worked in harmony to automate feeding and environmental monitoring, with a backend handling data flow and a mobile app offering user interaction.

Project highlights:

- **Communication Architecture:** ESP32-server-mobile communication was stable and responsive using TCP and WebSocket protocols.
- **Reliable Automation:** Feeding and environment sensing operated accurately under testing.
- **Mobile Control:** Flutter app allowed full system management with notifications.

References

[1] C., K. K., K. Subedi, S. Sharma, and P. Paneru (2024). Iot based smart poultry management system. Journal of IoT in Social, Mobile, Analytics, and Cloud 6(1), 39–53.

[2] Himel, G. M. S., M. M. Islam, M. N. Kader, and M. Rahman (2025). Artificial intelligence-based smart galliformes farm management system. Journal of Umm Al-Qura University for Engineering and Architecture 16, 64–77.